

Python collections.deque Methods Cheat Sheet

Method	1-Line Definition	Example	Time / Space
append(x)	Add element to the right end of deque.	d=deque([1,2]); d.append(3) → deque([1,2,3])	O(1) / O(1)
appendleft(x)	Add element to the left end of deque.	d=deque([2,3]); d.appendleft(1) → deque([1,2,3])	O(1) / O(1)
pop()	Remove and return the right-most element.	deque([1,2,3]).pop() → 3	O(1) / O(1)
popleft()	Remove and return the left-most element.	deque([1,2,3]).popleft() → 1	O(1) / O(1)
extend(iterable)	Add multiple elements to the right side.	d=deque([1]); d.extend([2,3]) → deque([1,2,3])	O(k) / O(k)
extendleft(iterable)	Add multiple elements to the left side (reversed order).	d=deque([3]); d.extendleft([2,1]) → deque([1,2,3])	O(k) / O(k)
clear()	Remove all elements from the deque.	d=deque([1,2]); d.clear() → deque([])	O(n) / O(1)
copy()	Create a shallow copy of the deque.	deque([1,2]).copy()	O(n) / O(n)
count(x)	Count occurrences of a value.	deque([1,2,1]).count(1) → 2	O(n) / O(1)
index(x,start,stop)	Return first position of a value.	deque([4,5,6]).index(5) → 1	O(n) / O(1)
insert(i,x)	Insert an element at a given position.	d=deque([1,3]); d.insert(1,2) → deque([1,2,3])	O(n) / O(1)
remove(x)	Remove first occurrence of a value.	d=deque([1,2,1]); d.remove(1) → deque([2,1])	O(n) / O(1)
reverse()	Reverse deque in place.	d=deque([1,2,3]); d.reverse() → deque([3,2,1])	O(n) / O(1)
rotate(n=1)	Rotate elements right by n (left if negative).	d=deque([1,2,3,4]); d.rotate(1) → deque([4,1,2,3])	O(min(n,len)) / O(1)
maxlen	Read-only attribute showing maximum size.	deque([1,2],maxlen=5).maxlen → 5	O(1) / O(1)

Deque Special Operations

- len(d) – Number of elements. Time: O(1)
- x in d – Membership test. Time: O(n)
- d[i] – Index access. O(1) near ends, O(n) worst-case in middle.
- d[0] – First element. O(1)
- d[-1] – Last element. O(1)

Notes

- deque stands for “double-ended queue”.
- append(), appendleft(), pop(), and popleft() are the primary reasons deque is preferred over list for queue operations.
- Unlike lists, inserting/removing at either end is O(1).